MSc Artificial Intelligence
Master Thesis

Master AI thesis template: A tale of computers

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Abstract

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Contents

List of Figures

List of Tables

List of Symbols

This list of notation can help the reader decipher the thesis:

$\mathbb{R}_+, \mathbb{R}_+^*, \mathbb{R}_-^*$	set of real numbers $\geq 0, > 0, < 0$, respectively;
$a \in \mathbb{R}$	scalar containing the value of the Answer.

List of Abbreviations

A list of acronyms could similarly be defined

CNN	Convolutional neural network
CPU	Central processing unit
SBU	System Billing Unit
SGD	Student Gradient Descent

Chapter 1

Introduction

Write your introduction here. It should be immediately clear how your proposed contribution is scientifically relevant and fills the research gap. This is a test citation Gruber (1995)

Towards the end of the introduction, you should also add your *preliminary* **reasearch questions (RQ)** here. You may want to state your main RQ like this:

To what extent can a master thesis template enhance the quality of the final thesis?

You can then list the sub-questions as:

- How does the structure of the template influence the final grading?
- To what extent is textual guidance sufficient for structured working?
- …

The double-column template page limit is 35 pages, 40 pages in single column. In any case, the irrelevant and the long unclear explanations will be penalized: clarity does not require long blocks of text, quite the opposite.

1.1 Section

1.1.1 Subsection

Subsubsection

Paragraph Ex culpa ut commodo proident esse excepteur mollit in sed aliqua fugiat sed cupidatat dolor sint quis veniam amet aute ea fugiat quis do excepteur sunt commodo magna cupidatat veniam minim anim mollit do enim et nostrud ad esse velit nisi est dolor do ut dolore excepteur nulla ex sunt exercitation consequat ullamco ad in fugiat ut ut enim adipisicing cillum nisi anim ullamco ullamco nisi esse aute dolore amet nisi adipisicing dolor dolore do ut aliqua in sunt aliqua tempor duis amet ut non adipisicing nostrud ex ullamco excepteur culpa anim est minim fugiat laborum ex irure consectetur eu in reprehenderit sint magna ut consequat laborum ex minim laboris culpa amet incididunt laborum laborum dolore nulla cupidatat dolore in aliqua ut eu in magna sunt eiusmod est labore sit eu incididunt fugiat in culpa pariatur voluptate duis id velit sunt reprehenderit pariatur amet esse dolore minim non fugiat fugiat amet incididunt commodo sit dolor.

In `itemize` and `enumeration`, longer sentences and text in paragraphs could require manual hyphenation with `-` in case of boxes under- or over-flows:

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2. Aliquip sunt in voluptate occaecat ut magna cupidatat sunt duis ut proident consequat.

Chapter 2

Related Work

Write about your related work here. Make clear to which key papers you will compare your eventual results. This can be done from the perspective of methods used, the task at hand and the addressed domain.

Remember: a related works section is not a "laundry list" of references, cited for the sake of citing papers: they are the support and motivation for your research gap and research question that you are going to address.

If you stick with Use `\cite{}` for reference that is part of the sentence, and `\citep{}` for references in parenthesis. For example, Viola and Wells III (1997) is awesome. Also, this is a citation (Viola and Wells III, 1997).

Chapter 3

Methodology

Write about your methodology here. Focus on your own contribution. Indicate exactly how you will assess your work in terms of evaluation.

3.1 Equations

We estimate the deformable template parameters θ_t and the deformation fields for every data point using maximum likelihood. Letting $\mathcal{V} = \{\boldsymbol{v}_i\}$ and $\mathcal{A} = \{a_i\}$,

$$\begin{aligned} \hat{\theta}_t, \hat{\mathcal{V}} &= \arg\max_{\theta_t, \mathcal{V}} \log p_{\theta_t}(\mathcal{V}|\mathcal{X}, \mathcal{A}) \\ &= \arg\max_{\theta_t, \mathcal{V}} \log p_{\theta_t}(\mathcal{X}|\mathcal{V}; \mathcal{A}) + \log p(\mathcal{V}), \end{aligned} \tag{3.1}$$

where the first term captures the likelihood of the data and deformations, and the second term controls a prior over the deformation fields.

Proof. Awesome proof. □

3.2 Long equations in two columns

Note that equations can fill the width pretty quickly when there are two columns. Different mathemtical environments, beyond the basic `\begin{equation}` may be of use. Taking as an example the binomial formula which overflows in Eq. (3.2):

$$(1+x)^n = \underbrace{(1+x) \times ... \times (1+x)}_{n \text{ times}} = \sum_{k=0}^{n} \binom{n}{k} x^k = \sum_{k=0}^{n} \frac{n!}{k!(n-k)!} x^k. \tag{3.2}$$

The `\begin{multline}` environment is an easy although crude fix:

$$\begin{multline} (1+x)^n = \underbrace{(1+x) \times ... \times (1+x)}_{n \text{ times}} \\ = \sum_{k=0}^{n} \binom{n}{k} x^k \\ = \sum_{k=0}^{n} \frac{n!}{k!(n-k)!} x^k. \end{multline} \tag{3.3}$$

`\begin{align}` tends to give the most control over the final result:

$$\begin{aligned} (1+x)^n &= \underbrace{(1+x) \times ... \times (1+x)}_{n \text{ times}} \\ &= \sum_{k=0}^{n} \binom{n}{k} x^k \\ &= \sum_{k=0}^{n} \frac{n!}{k!(n-k)!} x^k. \end{aligned} \tag{3.4}$$

3.3 Math styles

Different font styles can be used for equations:

- `$a b c A B C 1 2 3$`: $abcABC123$
- `$\mathbf{a b c A B C 1 2 3}$`: $\mathbf{abcABC123}$
- `$\mathfrak{a b c A B C 1 2 3}$`: $\mathfrak{abcABC123}$
- `$\mathcal{ABC}$`: $\mathcal{ABC}$
- `$\mathbb{ABC}$`: $\mathbb{ABC}$
- `$\mathsf{ABC}$`: ABC
- `$\mathtt{ABC}$`: $\mathtt{ABC}$

Text and names in equations should be dealt with the `\text` command, for instance:
`$\mathcal L_{\text{SuperLoss}}$`: $\mathcal{L}_{\text{SuperLoss}}$ and not $\mathcal{L}_{SuperLoss}$.

Chapter 4

Results

Write about your results here. Good captions to tables and/or figures are key.

4.1 Table and Figures

Table 4.1: By convention, Table caption goes on top.

Left	center	right
111	222	333
444	555	666

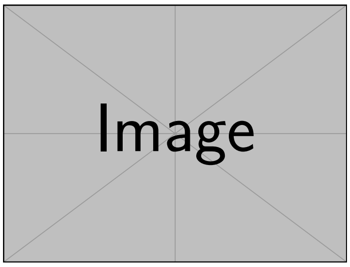

Figure 4.1: Example figure. Notice that the caption goes below and that there is a way to have a shorter caption for the list of Figures.

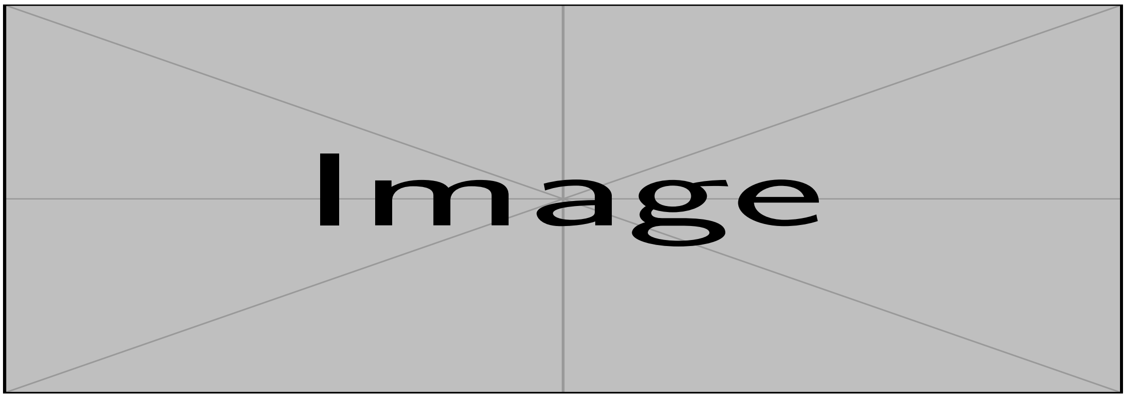

Figure 4.2: Example figure spanning the two columns.

Chapter 5

Discussion

Write your discussion here. Do not forget to use sub-sections. Normally, the discussion starts with comparing your results to other studies as precisely as possible. The limitations should be reflected upon in terms such as reproducibility, scalability, generalizability, reliability and validity. It is also important to mention ethical concerns.

Chapter 6

Conclusion

Write your conclusion here. Be sure that the relation between the research gap and your contribution is clear. Be honest about how limitations in the study qualify the answer on the research question.

Bibliography

Thomas R. Gruber. 1995. Toward principles for the design of ontologies used for knowledge sharing? *International Journal of Human-Computer Studies* 43, 5-6 (1995), 907–928. doi:10.1006/ijhc.1995.1081

Paul Viola and William M Wells III. 1997. Alignment by maximization of mutual information. *International journal of computer vision* 24, 2 (1997), 137–154.

Appendix A

First Appendix

Put your appendices here. Remember that a work and thesis should be readable and understandable *even without* reading the Appendix. Treat it as a "if you want to know more, read this" part.

A.1 Appendix section

A.1.1 Appendix subsection

Appendix subsubsection

Appendix paragraph Lorem ipsum dolor sit amet, consectetur adipisicing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.